

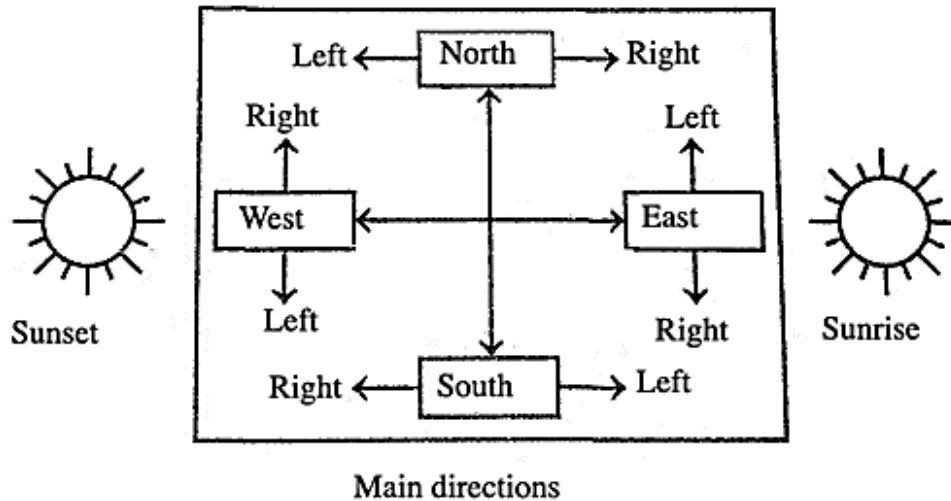
Distance and Direction Sense Test

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Direction is the measurement of position of one object with respect to another object and displacement is the minimum distance between the starting point and final point of an object.

Main Directions

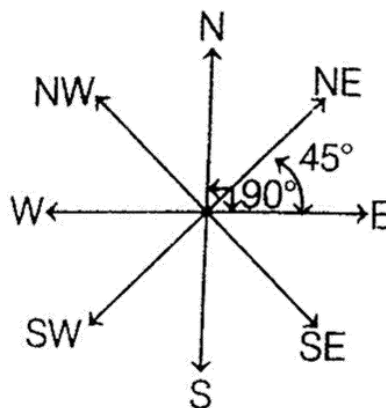
There are four main directions viz. East, West, North and South as shown in the figure given below



Abbreviations for these directions are E (East), W (West), N (North) and S (South).

Cardinal Directions

A direction between two main directions is called cardinal direction or sub-direction, i.e., NE (North-East), NW (North-West), SE (South-East) and SW (South-West).



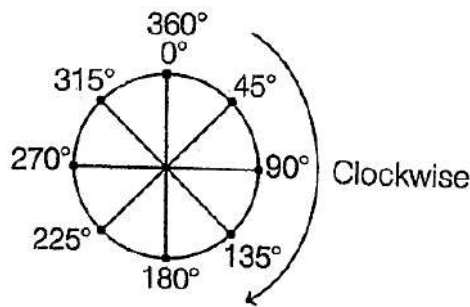
Students are advised to use diagram as given here, for the purpose of sensing directions.

Angle formed between two main directions (i.e., North-East; North-West; South-East or South-West) is 90° and angle formed between a cardinal direction and main direction is 45° .

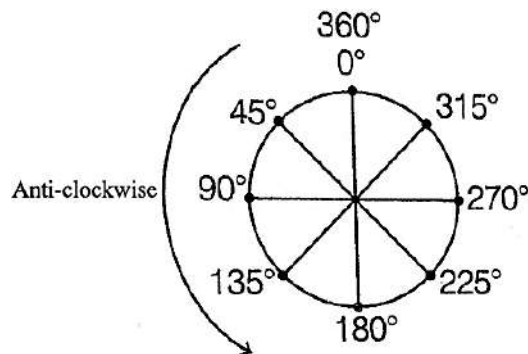
Angle of Movement

For solving questions based on angle of movement, it is necessary to know the rotations which are given below

(i) Movement towards the right is called clockwise (CW) movement.

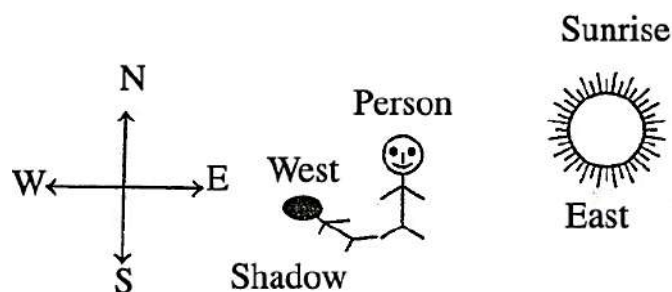


(ii) Movement towards the left is called anti-clockwise (ACW) movement.



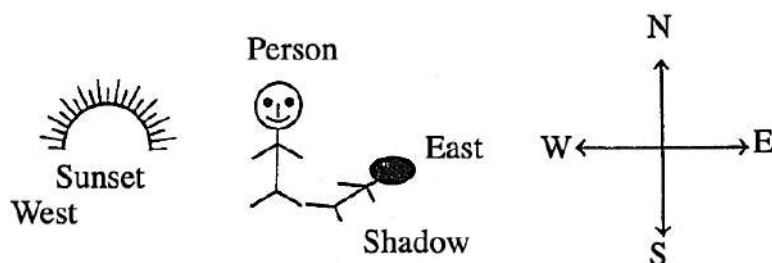
Shadow at the Time of Sunrise

In the morning, sunrise in the East, the shadow of any person or object falls in the West.



Shadow at the Time of Sunset

In the evening, when the sunset in the West, the shadow of a person or an object falls in the East.



Shortest Distance

If we draw a straight line from the initial point of the object to the final point, then length of this line is called shortest distance.

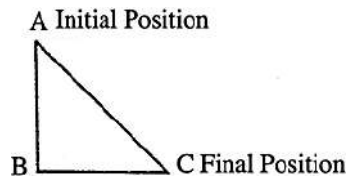
e.g., An object starts from point A and reaches to point C after going through point B.

Then, **distance between initial and final position = AB + BC**

But shortest distance between initial and final position = AC

To find the shortest distance between two points, it is necessary to know Pythagoras theorem.

Here, in ΔABC



AB = Perpendicular, BC = Base and AC = Hypotenuse. Hence, the shortest distance between A and C

i.e., $AC = \sqrt{AB^2 + BC^2}$.

Example 1: Ravi walks 8 km North-East and then 6 km South-East. Find the total distance as well as shortest distance between starting and end points.

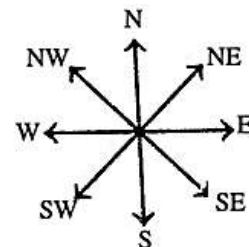
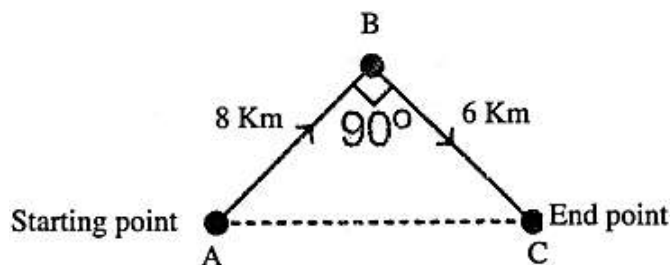
A. 14 km, 12 km

B. 14 km, 10 km

C. 10 km, 8 km

D. 8 km, 6 km

Solution: (B) According to the question,



Here, total distance = $AB + BC = 8 + 6 = 14$ km

and shortest distance $AC = \sqrt{AB^2 + BC^2} = \sqrt{8^2 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10$ km.